

Thowfiqur Bari Chowdhury

Final-Year B.Sc. Student in Computer Science & Engineering | Sylhet, Bangladesh
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Research Interests

Retrieval-Augmented Generation (RAG), Agentic AI, LLM reliability and hallucination detection, Bengali NLP, applied AI/ML, and emerging Medical/Bio + AI/ML research.

Education

Shahjalal University of Science and Technology (SUST)
B.Sc. in Computer Science and Engineering, Batch CSE-21

Expected Graduation: 2027
CGPA: 3.44

Research Projects

Bengali Health-Advisory RAG with Hallucination-Safety Layer *Thesis Direction — Ongoing*

- Investigating citation-grounded Bengali public-health advisory using curated knowledge bases, retrieval-grounded generation, and post-hoc hallucination and answer verification.

NexusRAG

Ongoing Engineering Project

- Designing and building a RAG system with PDF ingestion, dense + BM25 hybrid retrieval, multi-query and parent-document retrieval, and cross-encoder reranking.
- Developing conversation memory, a retrieval-comparison dashboard, authentication, and containerized deployment using Python, LangChain, Gemini, FAISS/Chroma, and BM25.

AdaptiveRAG-Agent

Exploratory Project

- Investigating LLM-guided retrieval decisions, tool-use patterns, and query routing to improve retrieval efficiency and answer quality over static RAG pipelines.

Selected AI/ML Projects

Bengali Hallucination Detection Pipeline

Kaggle Competition

- Built an ensemble combining fine-tuned BanglaBERT-large, a Qwen3-14B LLM-as-judge, and dense passage retrieval; implemented LightGBM meta-stacking with bootstrap-tuned weighting, achieving a **0.803 competition score**.

Vehicle Detection on Bangladeshi CCTV Footage

Kaggle Competition

- Analyzed a 13-class dataset with tiny-object and severe class-imbalance challenges; trained YOLO11 detectors using high-resolution inputs, AdamW, and copy-paste augmentation.

DentalFlow — AI-Enabled Dental Clinic CRM

- Developed a full-stack dental clinic CRM and integrated Groq LLM-based lead analysis using FastAPI, PostgreSQL, React, and n8n.

Achievements & Competitions

- Bengali Hallucination Detection (Kaggle):** Achieved a 0.803 competition score with a BanglaBERT, LLM-judge, retrieval, and LightGBM ensemble.
- bKash/NSUCEC Datathon:** Built a churn-prediction pipeline for a **75M+ row** transactional dataset using Dask, LightGBM, XGBoost, CatBoost, and Optuna; finished among the top competitors.
- Additional Kaggle work:** Road Vision vehicle detection; BirdCLEF+ multi-label acoustic species identification; Bengali long-form ASR using faster-whisper.

Technical Skills

Languages	Python, C, C++, Java, JavaScript, TypeScript, SQL
ML/DL & NLP	PyTorch, TensorFlow, scikit-learn, Transformers, LightGBM, OpenCV, Bengali NLP, hallucination detection
AI/GenAI	LLMs, RAG, Agentic AI, LangChain, embeddings, vector search, BM25 hybrid retrieval, reranking, Gemini API
Engineering	FastAPI, React, Node.js, PostgreSQL, MongoDB, Docker, Git, Linux, Kaggle